Name:

Roll No:

CH-101: Fundamental Concepts & Applications of Chemistry

Quiz-1

1 September 2025

Answer all the questions.
You can have scientific calculator and one handwritten A4 sheet.

1. What does the product of $\Psi^*(x)\Psi(x)$ represent in quantum mechanics? [**0.5 Mark**]

   A. Probability density   B. Normalization constant
   C. Square of wavefunction's real part   D. Expectation value

   Ans:

2. The normalization condition of a wavefunction $\Psi(x)$ ensures that: [**0.5 Marks**]

   A. Energy is quantized   B. Wavefunction is finite everywhere
   C. Probability over all space equals to unity   D. Wavefunction is acceptable

   Ans:

3. Consider the operators $\hat{D} = \dfrac{d}{dx}$ and $\hat{D}^2 = \dfrac{d^2}{dx^2}$, and the functions $\{e^{ikx}, \sin(kx), \cos(kx), xe^{ikx}\}$. Which of the following statements is true? [**1 Marks**]

   A. $\hat{D}$ and $\hat{D}^2$ commute and all functions are simultaneous eigenfunctions of both
   B. $\hat{D}$ and $\hat{D}^2$ do not commute for all the functions
   C. Only e^{ikx} is a simultaneous eigenfunction of $\hat{D}$ and $\hat{D}^2$.
   D. xe^{ikx} is not an eigenfunction of either $\hat{D}$ or $\hat{D}^2$.

   Ans:

4. The commutator of $\hat{x}$ with the Hamiltonian $\hat{H}$, i.e., $[\hat{x},\hat{H}]$, is [**1 Marks**]

   A. 0   B. $\frac{i\hbar}{m}$ C. $\frac{-\hbar^2}{2m}\hat{p}_x$ D. $\frac{i\hbar}{m}\hat{p}_x$

   Hints: Here $\hat{H} = (\frac{\hat{p}_x^2}{2m}) + \hat{V}(x)$. *Following identity* $[\hat{A},\hat{B}\hat{C}]=\hat{B}[\hat{A},\hat{C}] + [\hat{A},\hat{B}]\hat{C}$ *would be useful to evaluate* $[\hat{x},\hat{H}]$

   Ans:

5. A particle is in a state $\phi = 2\psi_1 + 4\psi_2$, where ψ_1 and ψ_2 are eigenfunctions of Hamiltonian with eigenvalues $E_1 = 3$ eV and $E_2 = 5$ eV, respectively. Evaluate the average/expectation value of the energy. [**1 Marks**]

   A. 95 eV   B. 4.6 eV   C. 4.0 eV   D. 23 eV

   Ans:

6. A particle of mass m is confined in a 2D rectangular box with sides a and $\frac{a}{2}$. Which of the following n_x and n_y correspond to the lowest degenerate energy level? [**1 Mark**]

   A. (1,1) and (2,1) B. (2,1) and (1,2) C. (2,2) and (4,1) D. (3,2) and (6,1)

   Ans:

7. How does the average momentum $\langle \hat{p} \rangle$ for the particle in a 1D box change with increasing quantum number n? [**1 Mark**]

   A. It increases linearly with n B. It remains same for all n C. It increases as n^2
   D. It decreases with n

   Ans:

8. Consider a particle of mass m confined in 1-D, 2-D square, and 3-D cubic boxes. Let ΔE be the energy difference between the third and second excited states, (*ignoring degeneracies*). Which of the following statements is correct? [**1 Mark**]

   A. ΔE(1-D box) = ΔE(2-D box) = ΔE(3-D box)
   B. ΔE(1-D box) > ΔE(2-D box) > ΔE(3-D box)
   C. ΔE(1-D box) > ΔE(3-D box) > ΔE(2-D box)
   D. ΔE(1-D box) < ΔE(2-D box) < ΔE(3-D box)

   Ans:

9. A particle is in the first excited state of a 1-D infinite square well of length $2L$. What is the probability of finding the particle in the last quarter of the box (i.e. $x = 3L/2$ to $x = 2L$)? *Hint:* $\displaystyle\int \sin^2(kx)dx = \frac{x}{2} - \frac{\sin(2kx)}{4k}$ [**1 Mark**]

   A. 0.09 B. 0.21 C. 0.30 D. 0.25

   Ans:

10. Approximate phenanthrene (with 16 π electrons) as a 2-D square box and answer the following.

    (a) Identify the quantum numbers (n_x, n_y) and degeneracies of the HOMO and LUMO levels. [**1 Mark**]

    A. (2,3) and 2; (3,3) and 1 B. (3,2) and 1; (3,3) and 2
    C. (2,2) and 1; (3,2) and 2 D. (1,3) and 2; (2,3) and 1

    Ans:

    (b) Evaluate the energy of the second excited state in terms of $\frac{h^2}{8ma^2}$. [**1 Mark**]

    A. 3 B. 4 C. 5 D. 6

    Ans:

Rough work

CH101 - Quiz-1, 1st September 2025.

Answer Key:

1. $\psi^*\psi$ represents 'Probability density'

   Correct option is **A**

2. Normalisation condition of wave function $\psi(x)$ ensures that probability over all space equals to unity.

   Correct option is **C**

3. $\hat{D} = d/dx$ $\quad \hat{D}^2 = d^2/dx^2$

   while evaluating commutation, one should consider functions among given four functions.

   $d/dx\,(e^{ikx}) = ik \times e^{ikx}$ $\quad d^2/dx^2\,(e^{ikx}) = -k^2 e^{ikx}$.

   $\hat{D}$ and $\hat{D}^2$ commutes only when the function is e^{ikx}.

   $x e^{ikx}$ does not commute with D and $\hat{D}^2$.

   Correct option are **C & D**

4. $$[x, \hat{H}] = \left[x, \frac{P_x^2}{2M} + V(x)\right] = \underbrace{[x, V(x)]}_{0} + \left[x, \frac{P_x^2}{2M}\right]$$

   $$\left[x, \frac{P_x^2}{2M}\right] = \frac{1}{2M}[x, P_x^2] = \frac{1}{2M}[x, P_x P_x]$$

$$= \frac{1}{2m}\left[P_x[x, P_x] + [x, P_x]P_x\right]$$

$$= \frac{1}{2m}\left[P_x\, i\hbar + i\hbar P_x\right] \quad \text{Since } [x, P_x] = i\hbar$$

$$\frac{1}{2m}[x, P_x^2] = \frac{1}{2m} \times 2 i\hbar P_x = \frac{i\hbar \hat{P}_x}{\cancel{2}m}$$

Correct option is D

5. $\phi = 2\psi_1 + 4\psi_2$

$E_1 = 3\,eV \qquad E_2 = 5\,eV.$

$$\langle E\rangle = \frac{|2|^2 E_1 + |4^2| E_2}{|2|^2 + |4|^2} = \frac{4\times 3 + 16\times 5}{4+16} = \frac{92}{20} = \underline{\underline{4.6\,eV}}$$

Correct option is B

6. $\text{Energy} \propto \frac{n_x^2}{a^2} + \frac{n_y^2}{(a/2)^2} = \cancel{n_x^2 + 4n_y^2}\; n_x^2 + 4n_y^2$

$(2,2) \Rightarrow 2^2 + 4\times 2^2 = 4+16 = 20.$

$\cancel{(1,4) \Rightarrow 1^2 + 4\times 4^2 = 1+}$

$(4,1) \Rightarrow 4^2 + 4\times 1^2 = 16+4 = 20.$

Correct option is C

7. $\langle \hat{p} \rangle = 0$ for all the quantum number 'n', for particle in a 1D box.

Correct option is **B**

8. For particle in a 1D box,

$$\text{Energy} \propto n^2 .$$

$$\Delta E_{3,3} = 16 - 9 = 7$$

In 2D box,

$$\text{Energy} \propto n_x^2 + n_y^2 .$$

1,1 $\rightarrow$ G.S

(1,2) (2,1) $\rightarrow$ 1st ex. state

(2,2) $\longrightarrow$ 2nd E.S

(3,1) (1,3) $\rightarrow$ 3rd E.S.

$$\Delta E_{3,2} = (3^2+1^2) - (2^2+2^2)$$

$$= 10 - 8 = \underline{\underline{2}}$$

For 3D box,

$$E \propto n_x^2 + n_y^2 + n_z^2 .$$

1,1,1

(121) (211) (112)

(221) (212) (122) Second E.S.

(311) (131) (113) third E.S

$$\Delta E_{32} = (3^2+1^2+1^2) - (2^2+2^2+1^2)$$

$$= 11 - 9 = \underline{2}$$

Typo in the options, correct it ~~to~~ from $>$ or $<$ to $\geq$ or $\leq$

Correct options are **B & C**

9. For $n=2$, $\psi^*\psi = \frac{1}{L}\sin^2\frac{\pi x}{L}$

[Note that the boxlength is $2L$ not L]

$$P = \int_{3L/2}^{2L} \frac{1}{L}\sin^2\frac{\pi x}{L} = \frac{1}{L}\left[\frac{x}{2} - \frac{\sin\frac{2\pi x}{L}}{4\left(\frac{\pi}{L}\right)}\right]_{3L/2}^{2L}$$

$$= \frac{1}{4} = \underline{\underline{0.25}}$$

Correct option is D

10. a)

n_x n_y	E	deg	Electron fill
(1,1)	2	1	2
(1,2) (2,1)	5	2	4
(2,2)	8	1	2
(3,1) (1,3)	10	2	4
(3,2) (2,3)	13	2	4 → HOMO (deg 2)
(1,4) (4,1)	17	2	→ ~~HOMO~~ LUMO (deg 2)
(3,3)	18	1	

16 electrons.

Corrected option 'c' is (3,2) and 2 : (1,4) and 2

Correct option is 'c'

b) Second excited state is

$\psi_{32/23} \longrightarrow \psi_{33}$, $18-13=\underline{\underline{5}}$

Correct option is 'c'

CH-101: Fundamental Concepts & Applications of Chemistry

Mid-semester examination

25 September 2025

- The total marks for the paper are 32. You are required to **attempt questions totaling 30 marks**.
- You may **skip one of the 2-mark questions** if needed. All other questions must be attempted fully.
- You can bring **one (only one) full A4 sheet** with the contents related to **only physical chemistry.**
- Except scientific calculator, no other devices are allowed.

....

1. What function contains all the information about a quantum particle? How can this function be used to obtain all measurable quantities and their time dependence if the Hamiltonian is time-independent? How does this compare to the classical picture of a particle? **(2 Marks)**

2. A quantum system has two stationary states ψ_1 and ψ_2 with energies E_1 and E_2. The system can be prepared in two different cases: (i) Classical pure case, entirely in one state (ψ_1 or ψ_2) with probabilities $|c_1|^2$ and $|c_2|^2$ where $|c_1|^2+|c_2|^2 = 1$. (ii) Superposition case, where $\Psi(0) = c_1\psi_1 + c_2\psi_2$, $|c_1|^2 + |c_2|^2 = 1$.
   (a) Write the time-dependent wavefunction $\Psi(t)$ for both cases. **(2 Mark)**
   (b) Using the probability density $P(t) = |\Psi(t)|^2$, explain how the two cases can be distinguished. *Hint: Consider how the cross-term in $|\Psi(t)|^2$ differs in the two cases.* **(2 Marks)**

3. Explain the stability of benzene using the particle in a 1-D box with length L and particle on a ring with radius R models. Take $L = 2\pi R$ so that the box length equals to the circumference of the ring. Six π-electrons occupy the lowest energy levels in both systems. *Hint: Start from the energy expressions for both models and compare the total energies after filling 6 electrons.* **(4 Marks)**

4. For a particle in a 1D box, explain how the energy gaps and probability density change as the quantum number n and particle mass m become very large. How does the particle's behavior in such cases resemble that of a classical particle? **(2 Marks)**

5. A system of 10 non-interacting electrons is confined in a 2-D square box of side L. Suddenly, one side of the box is reduced to $L/2$, making the box rectangular. Explain how the energy of individual levels and the total energy of the system change. **(2 Marks)**

6. Given that the HOMO – LUMO energy gap of hexatriene is 4.3 eV, estimate the effective conjugation length L of hexatriene by modeling it as electrons moving in a 1-D box. Note that hexatriene has six π electrons. ($h = 6.626 \times 10^{-34}$ Js, $m = 9.109 \times 10^{-31}$ kg, and 1 eV $= 1.602 \times 10^{-19}$ J). **(3 Marks)**

7. A particle is constrained to move on the surface of a sphere and is described by the angular momentum quantum number $l = 2$.

   (a) Determine the total energy, the magnitude of the total angular momentum, and the possible values of the z-component of angular momentum. Comment on the x- and y-components of angular momentum. **(2 Marks)**

   (b) Explain how the possible orientations of angular momentum change as l becomes large, and how this approaches the classical limit. **(2 Marks)**

8. For the hydrogen 1s orbital with $\psi_{1s}(r) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$, the wave function is maximum at $r = 0$ (at the nucleus). However, the most probable distance of the electron in the 1s orbital is not at the nucleus. Explain why? **(2 Marks)**

9. The wave function of a hydrogen atom at time t=0 is,

$$\psi(r, 0) = \frac{1}{\sqrt{10}}\left(2\psi_{100} + \psi_{210} + \sqrt{2}\,\psi_{211} + \sqrt{3}\,\psi_{21-1}\right),$$

   where the subscripts indicate the quantum numbers n, l, m.

   (a) Calculate the expectation values of energy E and total angular momentum $\hat{L}^2$ for this system. *Hint: The energy of a hydrogen-like atom is* $E_n = -\frac{13.6\ eV}{n^2}$ **(3 Marks)**

   (b) If a measurement of $\hat{L}^2$ and $\hat{L}_z$ is made, what is the probability of finding the system with $l = 1$ and $m = 1$? **(1 Marks)**

10. The effective nuclear charge for a 3d electron in copper is .... **(1 Mark)**

11. a) Ca, Ba and Sr & b) Y, Sc and Lu: arrange them in ascending order of their atomic radii
    a) .... and b) .... **(1 Mark)**

12. Be^{2+}, Al^{3+} and Ti^{4+} and F^-, I^-, H_2O: using these ions make three stable compounds .... (apply HSAB concept) **(1 Mark)**

13. I_2 has a lower melting point compared to F_2: True or False, and due to ....
    (1 Mark)

14. $CuCl_2$ vs Cu_2Cl_2, which one has higher covalent character .... **(1 Mark)**

.... END ....

Course code: CH101
Quiz-2 (10 marks)

Part A: Inorganic Chemistry (5 marks)

1. Draw all possible stereoisomers of $[Mn(ox)_2(NH_3)_2]^+$ (Note: structure of "ox" has to be clearly drawn). **(1 mark)**

2. Write name of the following compounds: (a) $[Co(H_2O)_2(NH_3)_4]SO_4$; (b) $[(en)_2Co(OH)(NH_2)Co(en)_2]Cl_4$. **(1 mark)**

3. Calculate oxidation state of metal and CFSE in the complexes, $[Fe(CN)_6]^{3-}$ and $[Co(Cl)_4]^{2-}$ **(1 mark)**

4. Which of the following complexes would you expect to suffer from a Jahn-Teller distortion: $[Cr(I_6)]^{4-}$, $[Cr(CN)_6]^{4-}$, $[Co(F)_6]^{3-}$, and $[Mn(ox)_3]^{3-}$? **(1 mark)**

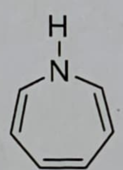
5. Write the electronic configuration and hybridization of central metal in: (a) $[Fe(CN)_6]^{4-}$; (b) $[Fe(H_2O)_6]^{2+}$. **(1 mark)**

Part B: Organic Chemistry (5 marks)

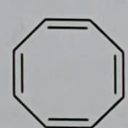
Objective type questions

(0.5x5=2.5 marks)

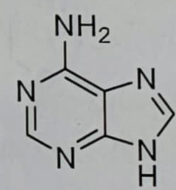
6. **Which of the following compounds are aromatic?**



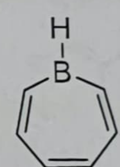
A



B



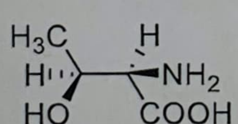
C



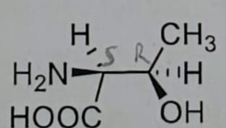
D

a) A, B, D b) C, B c) D, C d) A, C, D

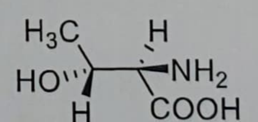
7. **Threonine, an amino acid, has four stereoisomers. The isomer found in nature is (2*S*,3*R*)-threonine. Which of the following structures represents this stereoisomer?**



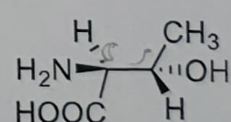
A



B



C



D

a) B b) C c) D d) A

8. **Which of the following statement (s) about fullerenes (C_{60}) are correct?**

m) They exhibit only substitution reactions

n) All the carbons are sp^2 hybridized

o) Sola rule can be used to explain their behavior

p) All the C-C bond lengths are equal

a) m and n

b) n and o

c) m and p

d) m, o, and p

9. **Ortho-chloroanisole upon reaction with $NaNH_2/NH_3$ will primarily form:**

a) Ortho-aminoanisole (o-anisidine)

b) meta-aminoanisole (m-anisidine)

c) para-aminoanisole (p-anisidine)

d) Aminobenzene (aniline)

10. **[8] annulene, in general, is a ____ molecule and removal of two electrons from it will make it ______.**

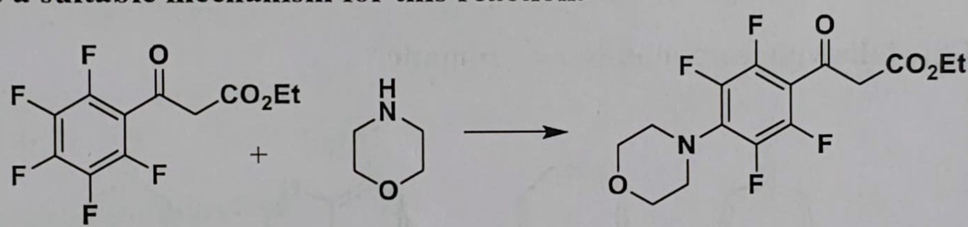
a) Antiaromatic, Aromatic

b) Nonaromatic, antiaromatic

c) Aromatic, antiaromatic

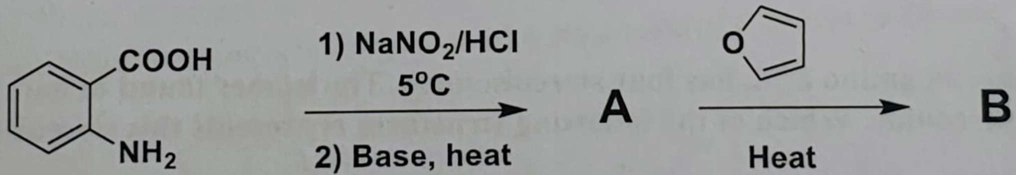
d) Nonaromatic, aromatic

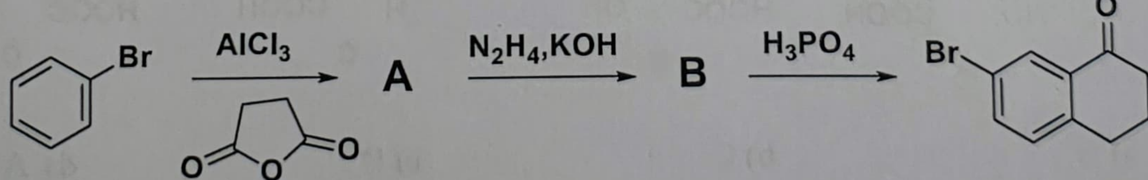
11. **Propose a suitable mechanism for this reaction.** **(1 mark)**



12. **Explain, using a structure, why azulene has a dipole moment of 1.8D.** **(0.5 mark)**

13. **Draw the structures of the missing products below.** **(0.25x4=1 mark)**





CH101: End semester examination (Total marks: 50)

Part A: Inorganic chemistry (Total marks: 23)

1. Calculate the magnetic moment for Gd^{3+} ($4f^7$) using L–S coupling. **(2 marks)**

2. The values of ε_{max} for the absorptions in the electronic spectra of $[CoCl_4]^{2-}$ and $[Co(OH_2)_6]^{2+}$ differ by a factor of about 100. Comment on this observation and state which complex among both you expect to exhibit the larger value of ε_{max}. **(2 marks)**

3. $[Et_4N]_2[NiBr_4]$ is paramagnetic, but $K_2[PdBr_4]$ is diamagnetic. Rationalize this observation. **(1 mark)**

4. For which member of the following pairs of complexes would Δ_{oct} be the larger and why: **(a)** $[Cr(OH_2)_6]^{2+}$ and $[Cr(OH_2)_6]^{3+}$; **(b)** $[Cr(F)_6]^{3-}$ and $[Cr(NH_3)_6]^{3+}$. **(2 marks)**

5. Draw the isomers of tris(ethylenediamine)cobalt(III) chloride (assign Δ and Λ). **(2 marks)**

6. Write at least one example of π-donor and π-acceptor ligands. State effect on Δ in octahedral complexes. **(3 marks)**

7. Compare atoms Si & Sn using HSAB and comment. **(2 marks)**

8. Define metallic bonding. **(1.5 marks)**

9. Arrange the following nano particles based on their melting temperature in the increasing order: 20 nm, 20 μm, 30 nm, 5 nm, 200 mm, 100 nm, 50 mm, 5 cm. **(2 marks)**

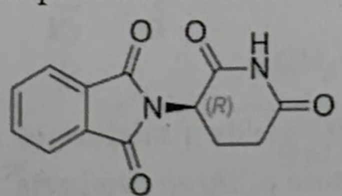
10. Match the size of gold nanoparticle with corresponding colour of the solution: Size of gold nanoparticle: 10 nm, 50 nm, 90 nm; Colour of solution: blue, red, violet **(2 marks)**

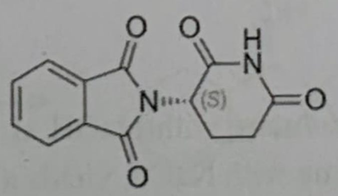
11. Define superparamagnetism. **(1.5 marks)**

12. Provide two reasons for unique properties of nanomaterials compared to bulk. **(2 marks)**

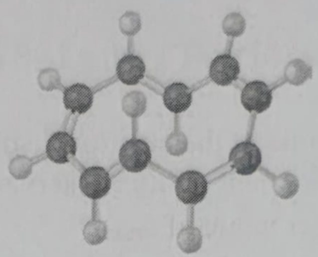
Part B: Organic chemistry (Total marks: 27) [Please be brief and to the point]

13. **(*R*)-thalidomide** is a drug (prevents nausea) while **(*S*)-thalidomide** is a teratogen (causes birth defects). These enantiomers usually interconvert in water which can be catalyzed by acidic conditions. Why? Propose a mechanism for the conversion of *R*-isomer to *S*-isomer. **(1.5)**

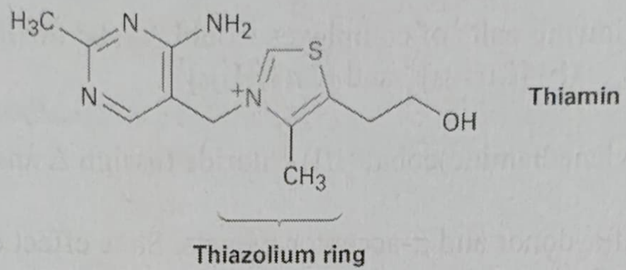




14. To be aromatic, a molecule must have $(4n+2)$ π electrons and must have a planar, monocyclic system of conjugation. **Cyclodecapentaene** fulfills one of these criteria but not the other and has resisted all attempts at synthesis. Explain. **(1)**



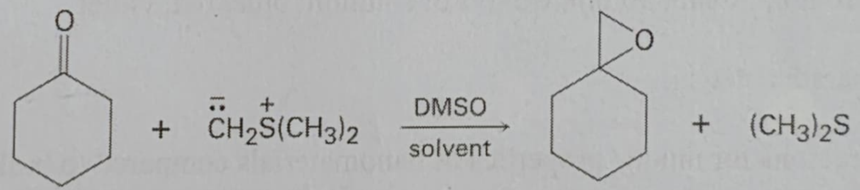
15. Thiamin (vitamin B1) contains a positively charged 5-membered nitrogen–sulfur heterocycle called a *thiazolium* ring. **How many aromatic rings does thiamin have**? Justify. **(1)**



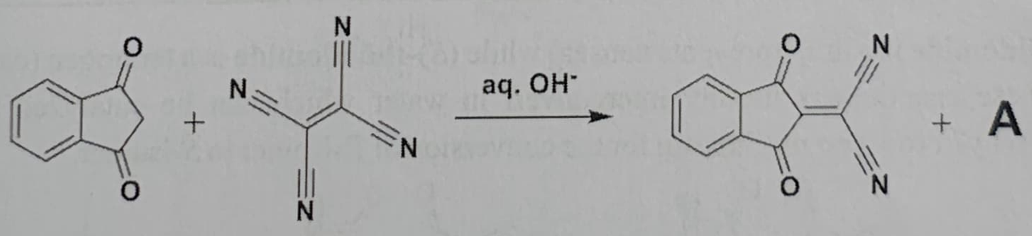
16. Draw MO diagrams (for each) to show which of the five molecular orbitals are occupied in **cyclopentadienyl cation, cyclopentadienyl radical, and cyclopentadienyl anion** respectively. Which, among these, is **aromatic** and why? **(1.5)**

17. The oxygen in water is primarily (99.8%) ^{16}O, but water enriched with the heavy isotope ^{18}O is also available. When an aldehyde/ketone is dissolved in **^{18}O-enriched water**, the isotopic label becomes incorporated into the carbonyl group. Explain with a mechanism. **(1.5)**

18. Ketones react with dimethylsulfonium methylide to yield **epoxides**. Rationalize. **(1)**

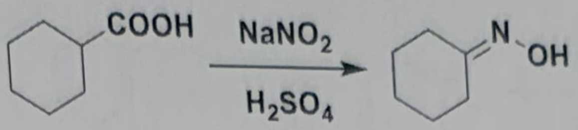


19. Rationalize the following reaction using a suitable mechanism and provide structure of A. **(2)**



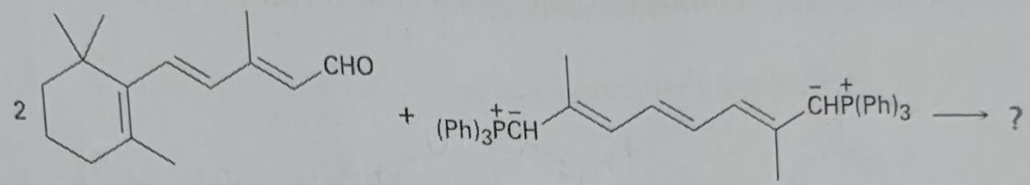
20. Treatment of **p-bromotoluene** with NaOH at 300°C yields a mixture of ***two*** **products**, but treatment of ***m*-bromotoluene** with NaOH yields a mixture of ***three*** **products**. Explain. **(1.5)**

21. Attempted nitrosation of this carboxylic acid leads to formation of **an oxime with loss of CO_2.** Explain using a suitable mechanism. (2)



22. This hydroxyketone shows no peaks in its **IR spectrum** around 1700 cm^{-1} (for carbonyl group), but it does show broad absorption around 3400 cm^{-1} (for hydroxyl group). Justify. (1)

23. ***β*-Carotene**, a yellow food-coloring agent and dietary source of vitamin A can be prepared by a *double* **Wittig reaction** between 2 equivalents of *β*-ionylideneacetaldehyde and a *diylide.* Show the structure of the *β*-carotene product with appropriate **stereochemistry.** (1)

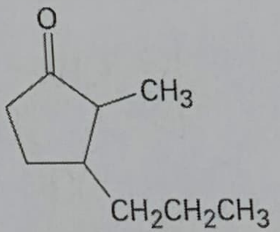


β-Ionylideneacetaldehyde

A diylide

24. Answer the following questions: (1x2=2)

a) How might you use a **Michael addition** reaction to prepare 2-methyl-3-propylcyclopentanone?



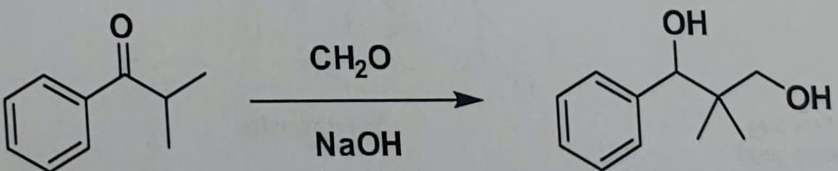
2-Methyl-3-propylcyclopentanone

b) What product would you expect from the following reaction? **Is this a named reaction?**

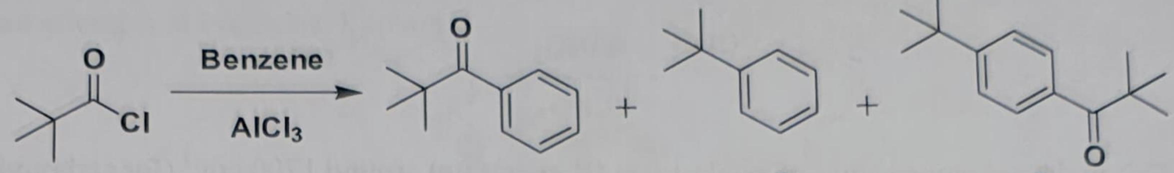
$$\mathrm{EtO\overset{\overset{O}{\|}}{C}CH_2CH_2\overset{\overset{CH_3}{|}}{C}HCH_2CH_2\overset{\overset{O}{\|}}{C}OEt} \xrightarrow[\text{2. } H_3O^+]{\text{1. } Na^+ \ ^-OEt} ?$$

25. Propose suitable mechanisms for the following reactions and identify the **name reaction.** (6)

a)

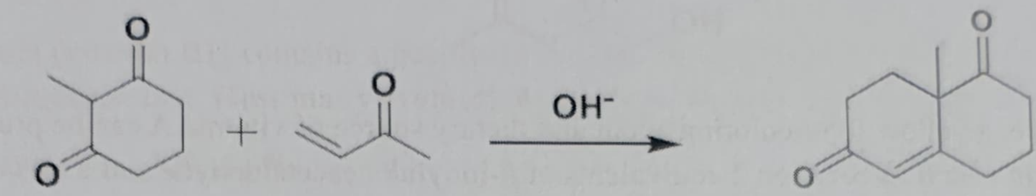


b)

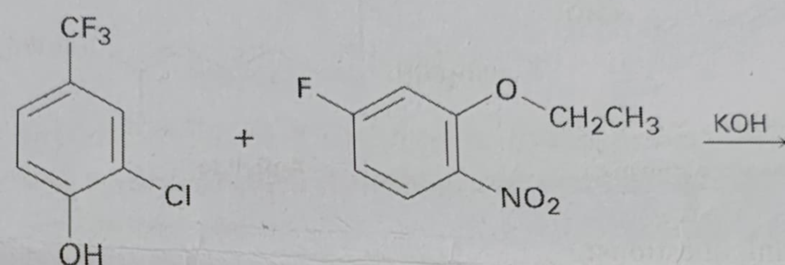


A B C

c)

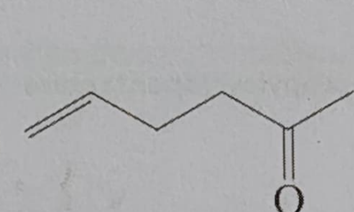


26. What is the product of the following reaction? **What kind of reaction is this?** (1)

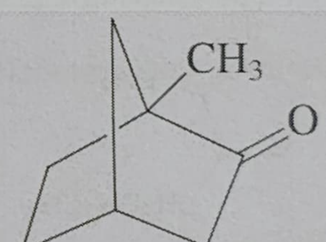


27. Predict the outcome of oxidation of the following compounds with a **peroxycarboxylic acid.** (1)

(a)

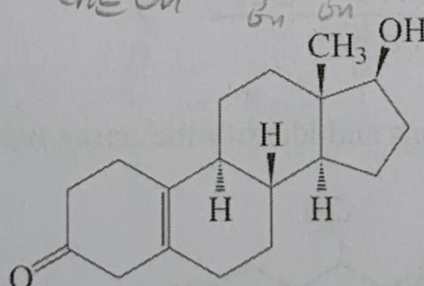


(b)

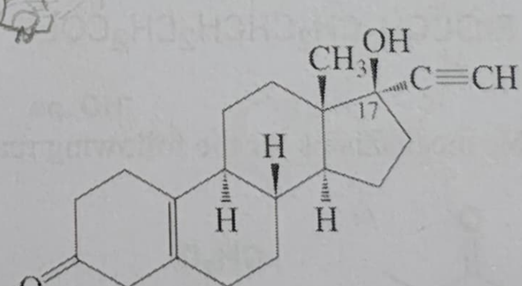


28. Propose a synthesis of **norethynodrel** using the starting compound (left-hand structure). [Hint: You can use acetylene, ethylene glycol, pyridinium chlorochromate as some of the reagents]. (2)

CH≡CH CH_2-CH_2 OH OH



Starting compound



Norethynodrel

① wave function $\psi(x,t)$ contains all the Information about a quantum particle : measurable quantities of the system can be obtained by using corresponding operators and wave function.

→ If the Hamiltonian is time - independent

$$\psi(x,t) = \sum_n c_n \psi_n(x) e^{-E_n t/\hbar}$$

→ In the case of classical particle, we need to know position and momentum of a particle.

② pure state ψ_1 or ψ_2 ; Super position $c_1\psi_1 + c_2\psi_2$.

ⓐ for ψ_1 $\quad \psi_1(t) = \psi_1(r)\, e^{-iE_1 t/\hbar}$

ψ_2 $\quad \psi_2(t) = \psi_2(r)\, e^{-iE_2 t/\hbar}$

for super positions $\psi(t) = c_1\psi_1(r)\, e^{-iE_1 t/\hbar} + c_2\psi_2(r)\, e^{-iE_2 t/\hbar}$

ⓑ probability $P(t) = |\psi(t)|^2$

for pure state $\quad P_{mix}(r,t) = |c_1|^2 |\psi_1(r)|^2 + |c_2|^2 |\psi_2(r)|^2$

for super position.

$$P_{sup}(r,t) = |c_1|^2 |\psi_1|^2 + |c_2|^2 |\psi_2(r)|^2 + c_1^* c_2 \psi_1^*(r)\psi_2(r)\, e^{-i(E_2-E_1)t/\hbar} + c_1 c_2^* \psi_2^*(r)\psi_1(r)\, e^{i(E_2-E_1)t/\hbar}.$$

The super position state has two additional cross terms, through which both the states can be distinguished.

③ $E_n(PIB) = \dfrac{n^2\hbar^2\pi^2}{2mL^2}$; $E_n(PoR) = \dfrac{m^2\hbar^2}{2mR^2}$ $L = 2\pi R$
$R = L/2\pi$.

$n = 1, 2, 3 \ldots$ $m = 0, \pm 1, \pm 2 \ldots$

$$E_n(PoR) = \frac{m^2\hbar^2}{2m\frac{L^2}{4\pi^2}} = \frac{4m^2\pi^2\hbar^2}{2mL^2}$$

Now fill six electrons in the levels of these boxes.

$n = 1, 2, 3$ levels are filled.

$$E_{PIB} = 2(E_1 + E_2 + E_3) = 2 \times \frac{\hbar^2\pi^2}{2mL^2}(1^2+2^2+3^2) = \frac{14\pi^2\hbar^2 \cancel{2}}{\cancel{2}mL^2} \times \cancel{2}$$

$$= \frac{14\pi^2\hbar^2}{2mL^2}$$

$$E_{Por} = 2(E_0 + E_1 + E_{-1}) = \cancel{2}\frac{4\hbar^2\pi^2}{\cancel{2}mL^2}(0^2+1^2+(-1)^2)$$

$$= \frac{8\hbar^2\pi^2}{mL^2}$$

$$\left(E_{Por} = \frac{8\hbar^2\pi^2}{mL^2}\right) < \left(E_{PIB} = \frac{14\pi^2\hbar^2}{mL^2}\right)$$

4. $$\Delta E_{PIB} = \frac{(2n+1)\pi^2\hbar^2}{2mL^2}$$ i.e $\Delta E \propto \frac{1}{m}$

→ with large mass "m" the Energy gap becomes smaller.

→ From the expression it looks like Energy gap ↑ses with ↑ses in 'n'. However the Individual energy ↑ses as n^2. So the relative Energy gap

$$\frac{\Delta E_n}{E_n} \propto \frac{2n+1}{n^2}.$$

∴ large n, the relative Energy spacing becomes very small

→ For large 'n' & large there will be huge number of levels within small region. Also, for large 'n' the probability density will have rapid oscillations because of these, the average probability density reaches a uniform distribution which is 11'ar for the consequence of small energy gap can be Interpreted as if all the Energy levels are allowed there is no discretization.

5. For square box $E_{n_x, n_y} = \frac{h^2}{8m}\left(\frac{n_x^2 + n_y^2}{L^2}\right)$

for $10\,e^-$ (1,1) (1,2) (2,1) (2,2) (1,3) are filled.

$$E_{tot} = 2(2+5+5+8+10)\frac{h^2}{8mL^2} = \frac{60\,h^2}{8mL^2}$$

for rectangular box $L_x = L/2$, $L_y = L$

$$E_{n_x n_y} = \frac{h^2}{8m}\left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2}\right) = \frac{h^2}{8m}\left(\frac{4n_x^2}{L^2} + \frac{n_y^2}{L^2}\right)$$

$$= 4n_x^2 + n_y^2\,(h^2/8mL^2)$$

$$E_{(1,2)} = 4+4\,(h^2/8mL^2) = \frac{8\,h^2}{8mL^2};$$

$$E_{(2,1)} = 16+1\,(h^2/8mL^2);$$

$$E_{tot} = 2(\underset{(1,1)}{5} + \underset{(1,2)}{8} + \underset{(2,1)}{17} + \underset{(2,2)}{20} + \underset{(1,3)}{13})\,h^2/8mL^2$$

$$= 126\,(h^2/8mL^2)$$

→ Note that ~~the~~ energetic order of levels in the rectangular box significantly changed as compared to the square box. i.e the order now is (1,1) (1,2) (1,3) (2,1) (2,2).

→ Individual Energies significantly ↑es when the box becomes square to rectangular.

→ Total Energy rises from 60 → 126.

→ Degeneracy gets lifted.

6] $E = \frac{n^2h^2}{8mL^2}$ For 6 electrons, till $n=3$ is filled.

So, $n=3$ is HOMO and $n=4$ is LUMO.

$$E_4 - E_3 = \frac{(4^2-3^2)h^2}{8mL^2} = \frac{7h^2}{8mL^2}.$$

ΔE is given as $4.3\,eV \rightarrow 4.3 \times 1.602 \times 10^{-19}\,J$

$= 6.89 \times 10^{-19}\,J.$

$$6.89 \times 10^{-19} = \frac{7h^2}{8mL^2}$$

$$L = \sqrt{\frac{7h^2}{8m \times 6.89 \times 10^{-19}}}$$

$$= \sqrt{\frac{7 \times (6.626 \times 10^{-34})^2}{8 \times 9.109 \times 10^{-31} \times 6.89 \times 10^{-19}}}$$

$$= \sqrt{\frac{3.073 \times 10^{-66}}{5.017 \times 10^{-48}}} = \sqrt{6.126 \times 10^{-19}} = 2.47 \times 10^{-9}\,m$$

$= \underline{2.47\,nm}.$

7] $E = \frac{L^2}{2I} = \frac{l(l+1)\hbar^2}{2I}$ for $l=2$, $E = \frac{6\hbar^2}{2I}$

magnitude of angular momentum $|L| = \sqrt{L^2}$

$= \sqrt{l(l+1)}\,\hbar = \sqrt{6}\,\hbar.$

Possible values of the Z-component of angular momentum

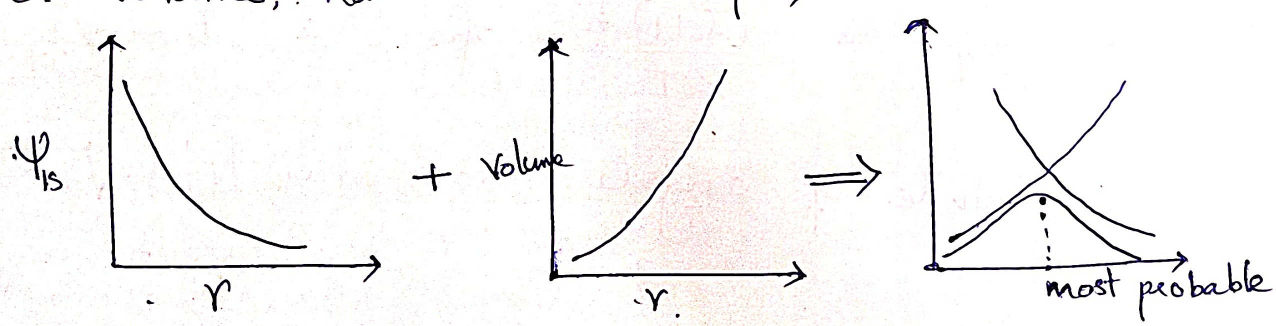
$L_z = m\hbar$ for $l=2$, $L_z = -2\hbar, -\hbar, 0, \hbar, 2\hbar$

→ L_x and L_y can not be determined along with L_z, because of the uncertainity principle. So there is huge uncertainity in angular momentum in x and Y axis.

For large 'l', huge number of L_z possible, ie, angular momentum vector can point in almost any direction, which approaches to classical limit.

8] $\psi_{1s} = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$.

From the form of function, the value is maximum at small 'r'. Accordingly, the probabulity of finding electron should be maximum at small 'r', or at the nucleus. But this is not the case. This can be explained as follows. There is another factor in the probabality at a fixed r, which is the volume. The volume increases with r. So with with r, the function and volume, have below shape,



So eventhough function value is maximum at small r, the volume is negligible. Therefore, the probabulity for small 'r' becomes very small.

9] (a) $\Psi(r, t=0) = \frac{1}{\sqrt{10}}\Big(\underset{\downarrow \atop l=0}{2\psi_{100}} + \underset{\downarrow \atop l=1}{\psi_{210}} + \underset{\downarrow \atop l=1}{\sqrt{2}\,\psi_{211}} + \underset{\downarrow \atop l=1}{\sqrt{3}\,\psi_{21-1}}\Big)$.

$$\langle E\rangle = \sum_i |c_i|^2 E_{ni} = |C_{100}|^2 E_1 + |C_{210}|^2 E_2 + |C_{211}|^2 E_2 + |C_{21-1}|^2 E_2.$$

$$= \frac{4}{10} E_1 + E_2\left(\frac{1}{10} + \frac{2}{10} + \frac{3}{10}\right)$$

$$= \frac{2}{5}(-13.6) + \frac{3}{5}(-3.4)$$

$$= 0.4(-13.6) + 0.6(-3.4)$$

$$\langle E\rangle = \underline{\underline{7.48\ eV}}.$$

Total angular momentum,

$$\langle L^2\rangle = \sum_i |c_i|^2 l_i(l_i+1)\hbar^2$$

$$= \frac{4}{10}\times 0 + \frac{1}{10}\times 2 + \frac{2}{10}\times 2\Big)\hbar^2$$

$$= \underline{\underline{1.2\hbar^2}}$$

(b) Probability of finding $l=1, m=1$ is $|C_{211}|^2$

$$= 2/10 = 1/5 = \underline{\underline{0.2}}$$

10] 7.85

11] (a) Ca < Ba < Sr

(b) Sc < Lu < Y

12] BeI_2, $Al(H_2O)_3$ and TiF_4

13] False, stronger temporary dipole-dipole interactions.

14] $CuCl_2$.

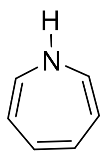
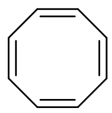
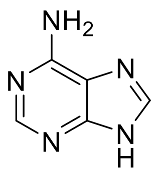
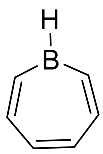
Course code: CH101
Quiz-2 (10 marks)

Part B: Organic Chemistry (5 marks)

Objective type questions

Correct answers are marked in red **(0.5x5=2.5 marks)**

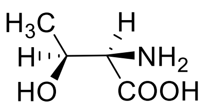
1. Which of the following compounds are aromatic?

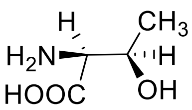





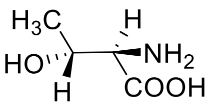
A B C D

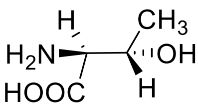
a) A, B, D **b)** C, B **c)** D, C **d)** A, C, D

2. Threonine, an amino acid, has four stereoisomers. The isomer found in nature is (*2S*,*3R*)-threonine. Which of the following structures represents this stereoisomer?


A


B


C


D

a) B **b)** C **c)** D **d)** A

3. Which of the following statement (s) about fullerenes (C_{60}) are correct?

m) They exhibit only substitution reactions n) All the carbons are sp^2 hybridized

o) Sola rule can be used to explain their behavior p) All the C-C bond lengths are equal

a) m and n **b)** n and o **c)** m and p **d)** m, o, and p

9. Ortho-chloroanisole upon reaction with $NaNH_2/NH_3$ will primarily form:

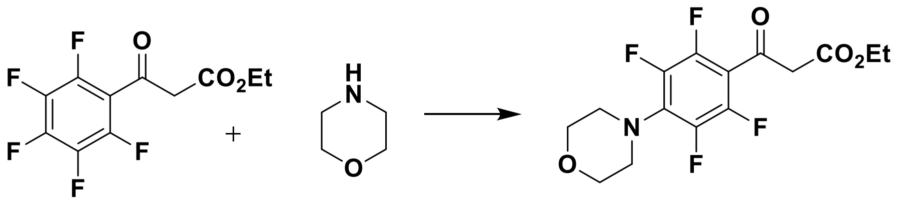
a) Ortho-aminoanisole (o-anisidine) **b)** meta-aminoanisole (m-anisidine)

c) para-aminoanisole (p-anisidine) **d)** Aminobenzene (aniline)

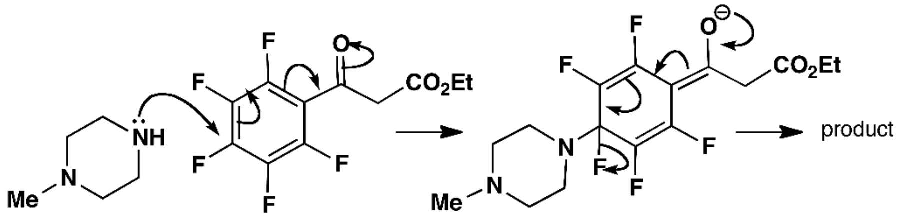
10. [8] annulene, in general, is a ____ molecule and removal of two electrons from it will make it ______.

a) Antiaromatic, Aromatic
b) Nonaromatic, antiaromatic
c) Aromatic, antiaromatic
d) Nonaromatic, aromatic

11. Propose a suitable mechanism for this reaction. **(1 mark)**

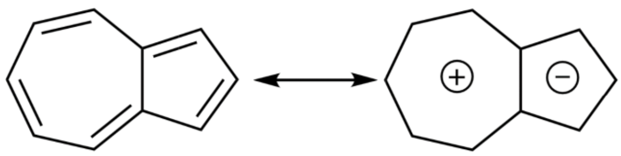


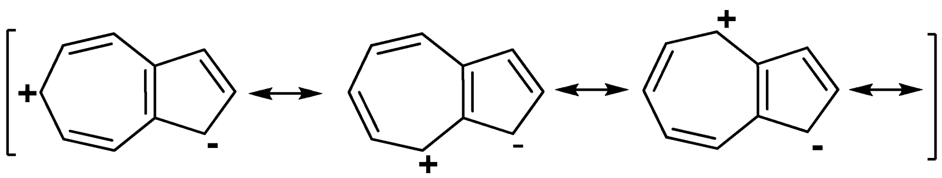
This is a nucleophilic aromatic substitution (S_NAr, addition-elimination mechanism), and major product is this p-substituted one. Structure of the nucleophile is different here (ignore N-Me group in the figure below), but mechanism remains the same. The attack will happen from N, as compared to O in the nucleophile (N is less electronegative and will easily donate electrons).

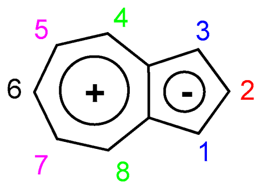


12. Explain, using a structure, why azulene has a dipole moment of 1.8D. **(0.5 mark)**

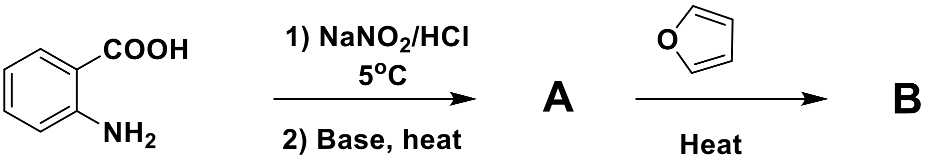
Electron delocalization in azulene will make one ring cycloheptatrienyl cation and the other ring cyclopentadienyl anion, both of which are aromatic. This makes charge separation in the compound, accounting for its dipolar character and the dipole moment.

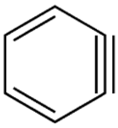


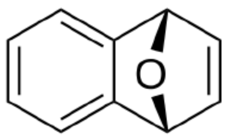


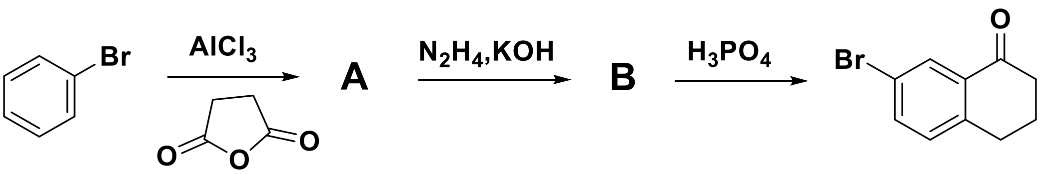


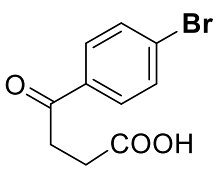
13. Draw the structures of the missing products below. **(0.25x4=1 mark)**



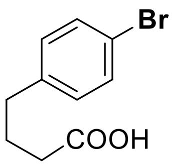
A: Benzyne 

B: Diels-Alder Product 





A



B

Friedel-Crafts acylation will give A, followed by Wolff-Kishner reaction will give B. (Mechanism is not required, each structure has 0.25 marks)

3) $Fe(CN)_6]^{3-}$ Oxidation state = +3

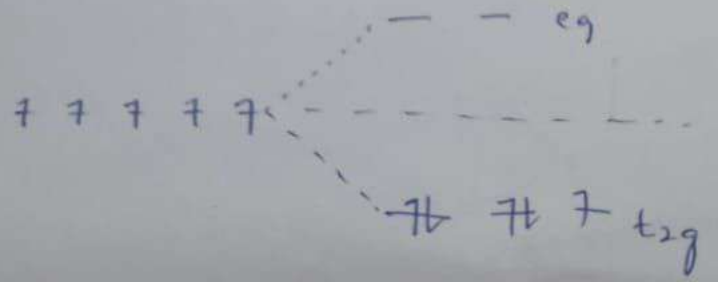
$Fe^{3+} = 4s^0 3d^5$

CFSE

∴ octahedral Complex

CN^- strong field ligand

pairing takes place

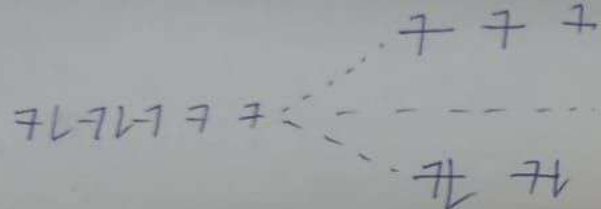


$CFSE = [-0.4(5) + 0.6(0)]\Delta_o$

$CFSE = -2.0 + 2P$

$[CoCl_4]^{2-} \Rightarrow Co^{2+} = d^7 \; 4s^0 3d^7$

tetrahedral Complex



$CFSE = [-0.6(4) + 0.4(3)]\Delta_t$

$[-2.4 + 1.2]\Delta_t$

$CFSE = -1.2\Delta_t$

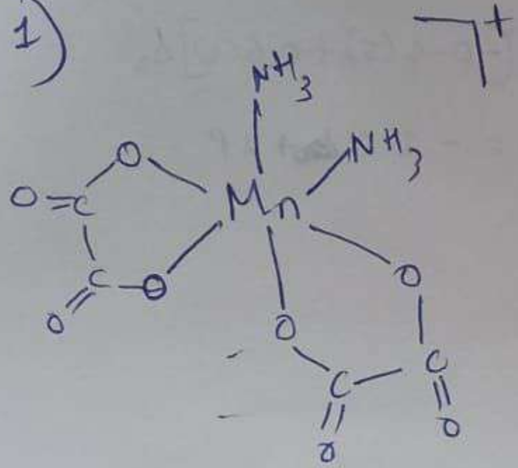
4) $[Cr(I_6)]^{-4}$ and $[Mn(ox)_3]^{-3}$

suffer from Jahn-Teller distortion

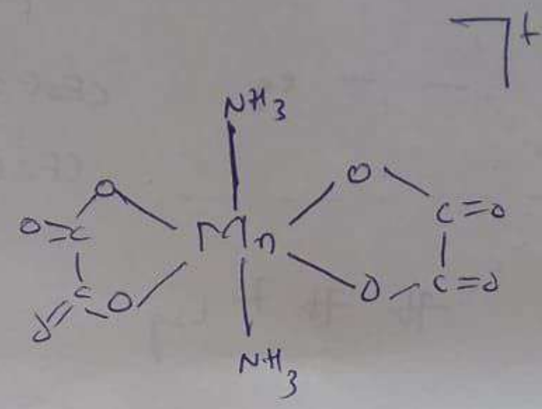
	Electronic configuration	Hybridization
(5) (a) $[Fe(CN)_6]^{-4} \Longrightarrow$	$3d^6$ or $3d^6 4s^0$	d^2sp^3
(b) $[Fe(H_2O)_6]^{+2} \Longrightarrow$	$3d^6$ or $3d^6 4s^0$	sp^3d^2

PART-A Inorganic

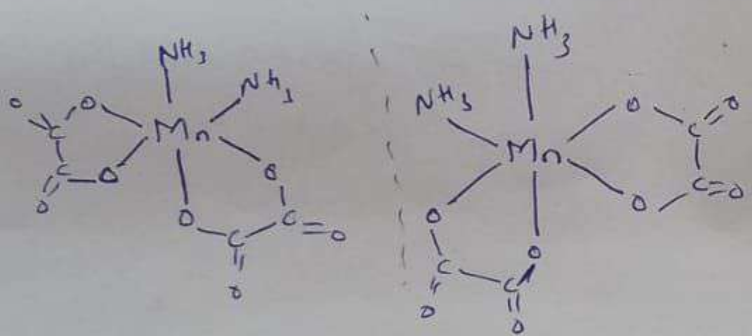
1)



cis



trans.



total <u>three</u> isomers
cis, trans,
cis is optically active.

2) $[Co(H_2O)_2(NH_3)_4]SO_4$
↳ tetraammine diaqua cobalt(II) sulphate

$[(en)_2Co(OH)_2(NH_2)Co(en)_2]Cl_4$

~~μ-amido~~ - μ-hydroxido ~~bis tetrakis (diethy amine) di cobalt (III)~~
~~Chloride~~

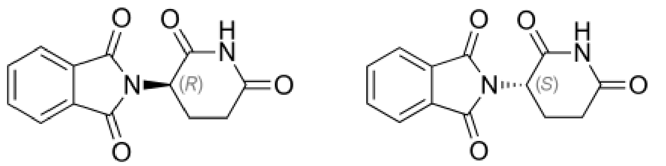
Bis(ethylenediamine) cobalt (III) μ-amido μ-hydroxo
bis(ethylenediamine) cobalt (III) Chloride .

μ-amido - μ-hydroxido di (ethylenediamine cobalt (III)) chloride
~~di ((bis(ethylenediamine) cobalt (III)) chloride~~

CH101: End semester examination (Total marks: 50)

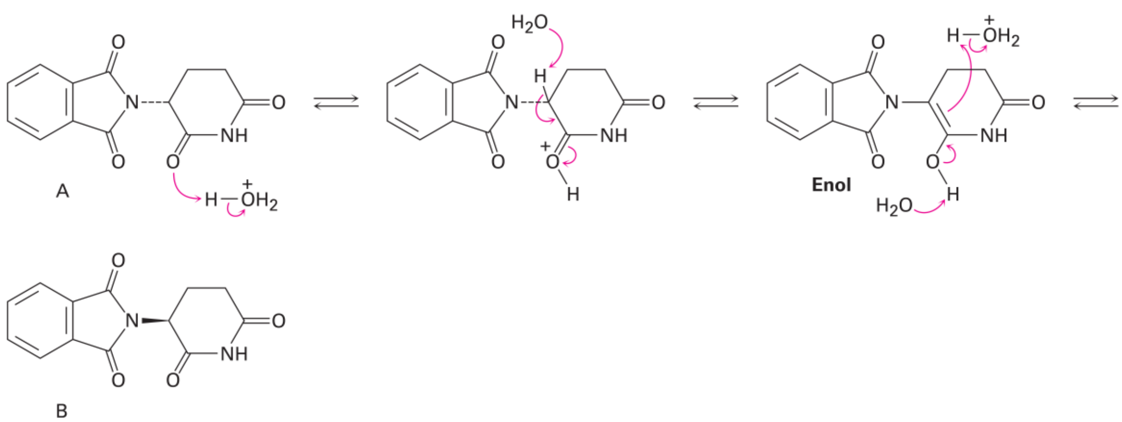
Part B: Organic chemistry (Total marks: 27) [Answers are in red]

13. (***R***)**-thalidomide** is a drug (prevents nausea) while (***S***)**-thalidomide** is a teratogen (causes birth defects). These enantiomers usually interconvert in water which can be catalyzed by acidic conditions. Why? Propose a mechanism for the conversion of *R*-isomer to *S*-isomer. **(1.5)**

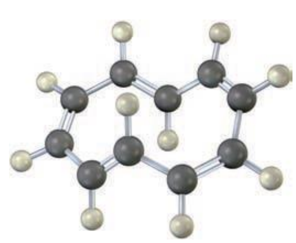


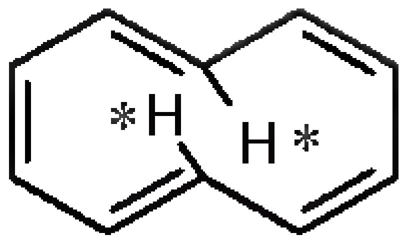
Here, the first enol formation occurs at the chiral center (assisted by the acid) of one isomer (say R) which converts the hybridization on the carbon from *sp*^{3} to *sp*^{2}. In the next step, upon going to the keto form, addition of hydrogen may happen on the top face, leading to the epimerized product (S-isomer). Enolization helps in the epimerization (similar mechanism to the one shown below, acid catalyzes the enolization).

5. Interconversion of the two forms of thalidomide takes place in an aqueous medium. This mechanism involves a type of rearrangement called epimerization (isomerization at a single chiral center).



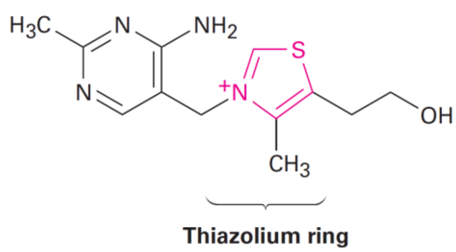
14. To be aromatic, a molecule must have $(4n+2)$ π electrons and must have a planar, monocyclic system of conjugation. **Cyclodecapentaene** fulfills one of these criteria but not the other and has resisted all attempts at synthesis. Explain. **(1)**





Cyclodecapentaene ([10] annulene) has (4n + 2) π electrons (n = 2), but it is not flat. If cyclodecapentaene were planar, the starred hydrogen atoms would crowd each other across the ring, resulting in Vanderwall's repulsions. To avoid this interaction, the ring system is distorted from planarity and the molecule becomes non-aromatic.

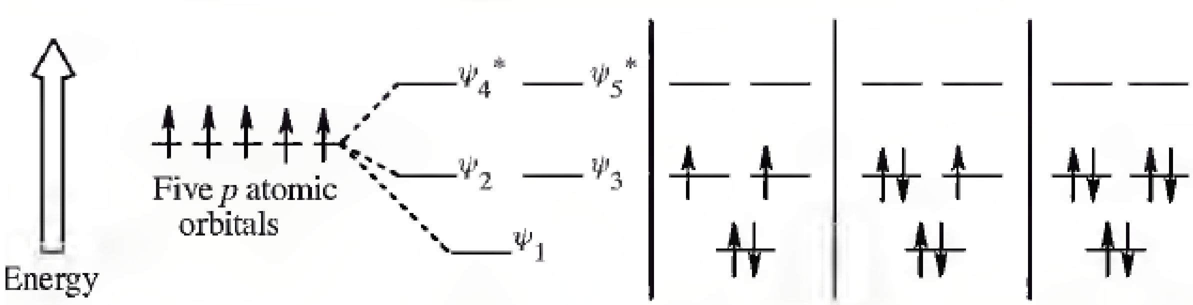
15. Thiamin (vitamin B1) contains a positively charged 5-membered nitrogen–sulfur heterocycle called a *thiazolium* ring. **How many aromatic rings does thiamin have**? Justify. **(1)**



Thiamin

It has two aromatic rings. In the pyrimidine ring on the left, lone pairs of the nitrogen atoms do not participate in resonance (6π electrons) while in the thiazolium ring, lone pair from sulfur atom participates in resonance (6π electrons). Since both rings are cyclic, planar, and conjugated and obey the Huckel rule, hence they both are aromatic.

16. Draw MO diagrams (for each) to show which of the five molecular orbitals are occupied in **cyclopentadienyl cation, cyclopentadienyl radical, and cyclopentadienyl anion** respectively. Which, among these, is **aromatic** and why? (**1.5**)

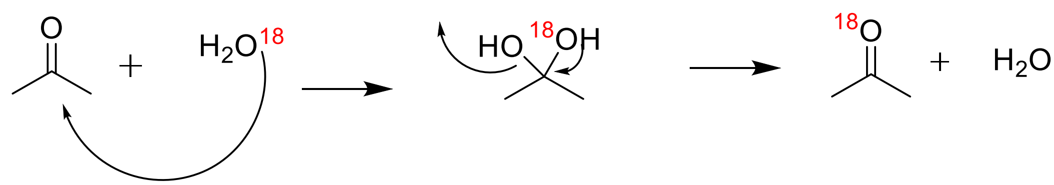


5 MOs **Cation (4 π)** **Radical (5π)** **Anion (6π)**

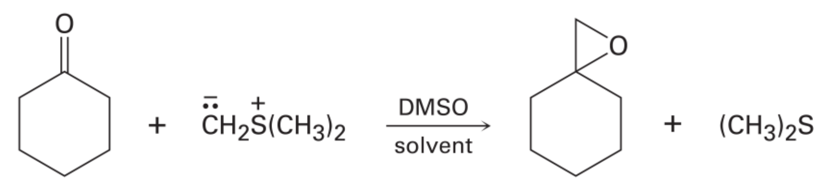
Since the anion obeys Huckel rule [4n + 2) π electrons], it is aromatic amongst the three.

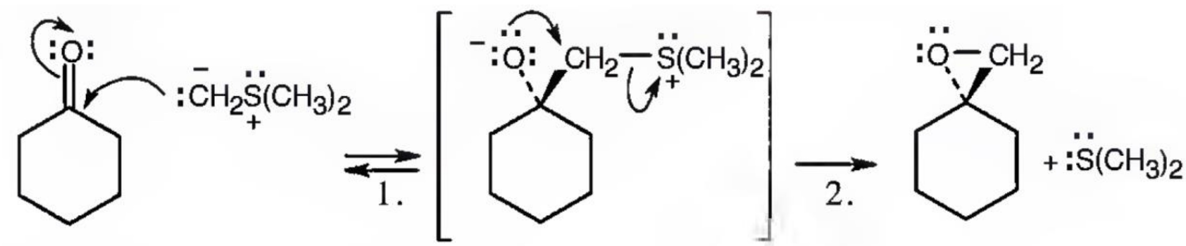
17. The oxygen in water is primarily (99.8%) ^{16}O, but water enriched with the heavy isotope ^{18}O is also available. When an aldehyde/ketone is dissolved in **^{18}O-enriched water**, the isotopic label becomes incorporated into the carbonyl group. Explain with a mechanism. **(1.5)**

This is just hydration (nucleophilic addition of water on C=O), followed by the reverse reaction which may lead to incorporation of the 18O isotope in alehede/ketone (here acetone is used as an example)



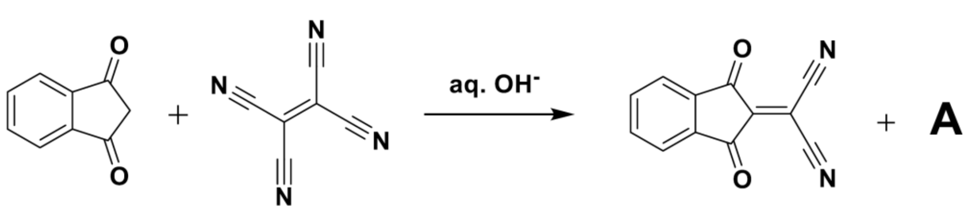
18. Ketones react with dimethylsulfonium methylide to yield **epoxides**. Rationalize. **(1)**



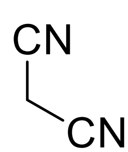


Step 1: Addition of ylide.
Step 2: S_N2 displacement of dimethyl sulfide by O^-.

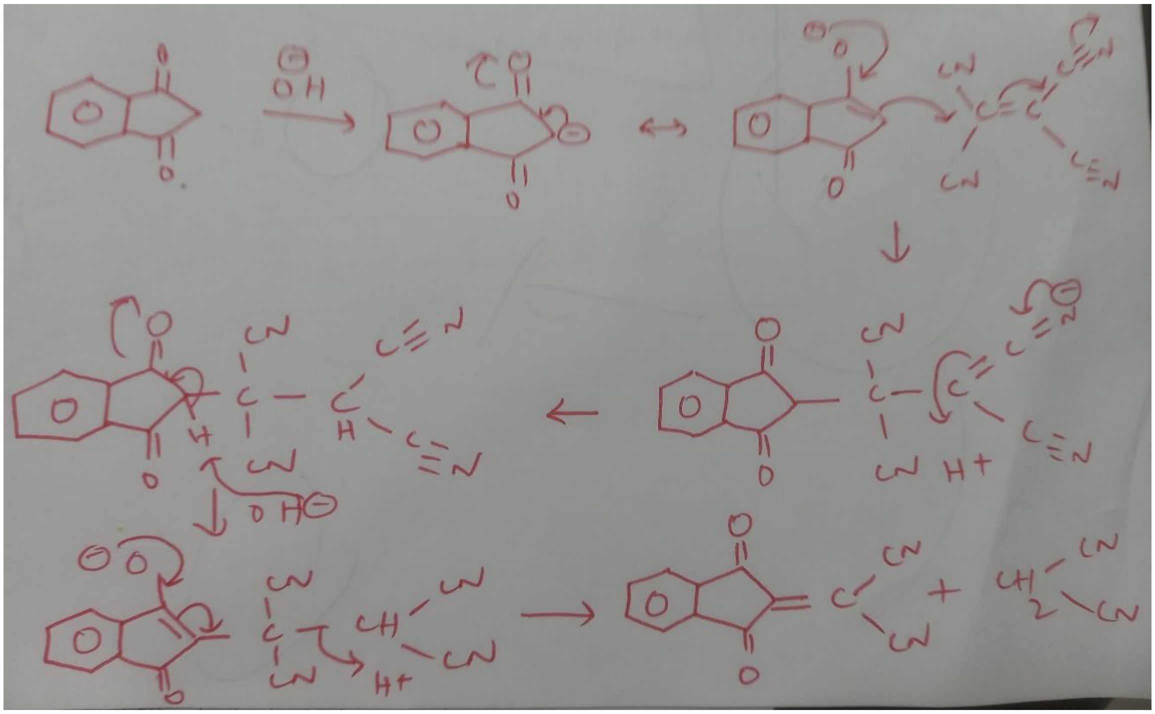
19. Rationalize the following reaction using a suitable mechanism and provide structure of **A. (2)**



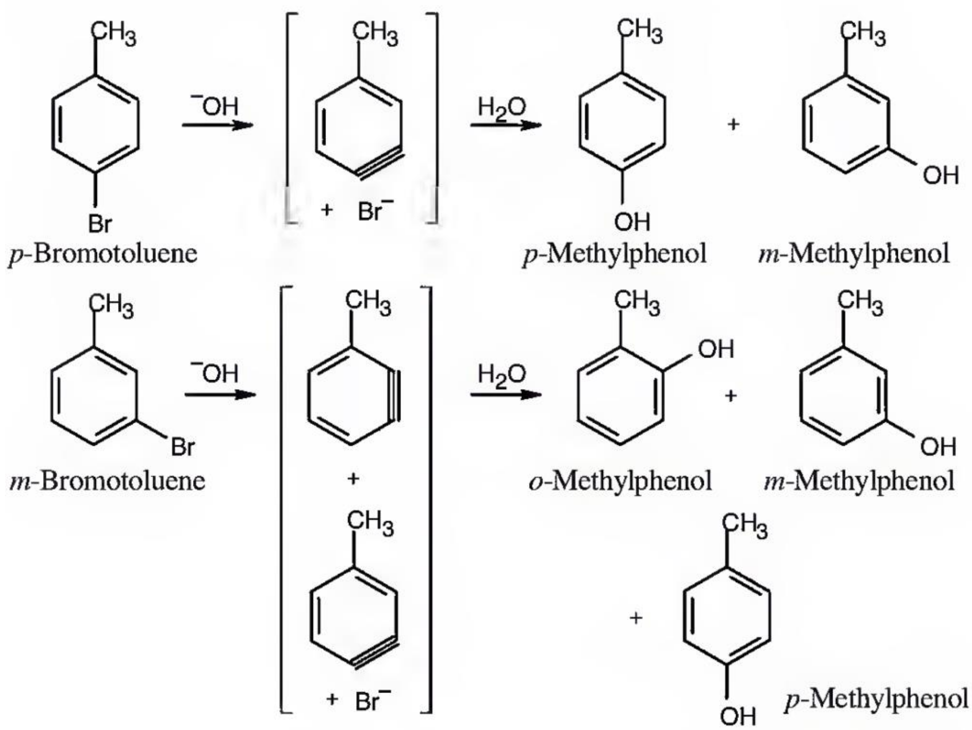
Structure of A



Mechanism involves conjugate addition

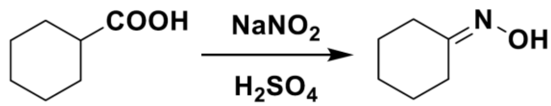


20. Treatment of ***p*-bromotoluene** with NaOH at 300°C yields a mixture of ***two* products**, but treatment of ***m*-bromotoluene** with NaOH yields a mixture of ***three* products**. Explain. **(1.5)**



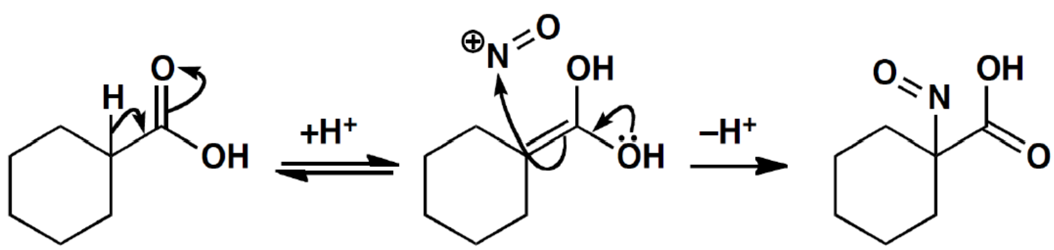
Treatment of *m*-bromotoluene with NaOH leads to two possible benzyne intermediates, which react with water to yield three methylphenol products.

21. Attempted nitrosation of this carboxylic acid leads to formation of **an oxime with loss of CO_2**. Explain using a suitable mechanism. **(2)**

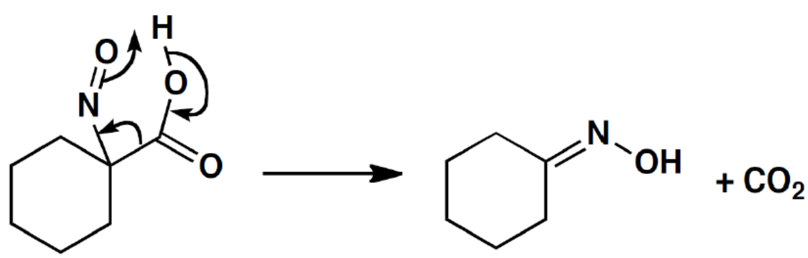


This will first involve enolization due to acid, followed by attack of NO+ and loss of CO2.

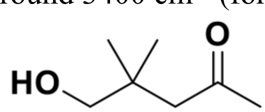
- First step: Enolization followed by reaction with NO+



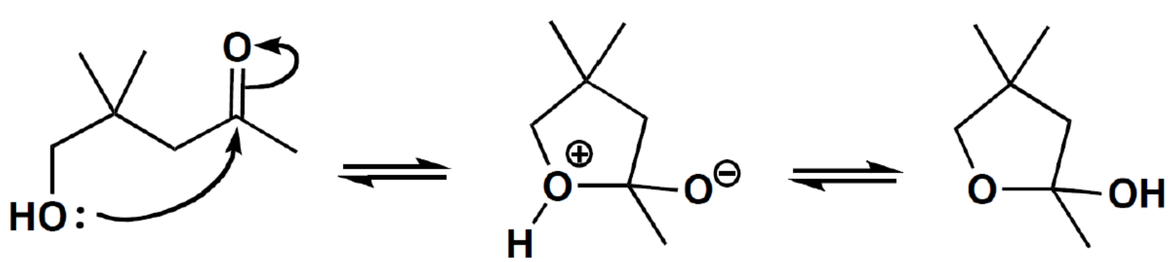
- 2nd step: Loss of CO2 and oxime formation



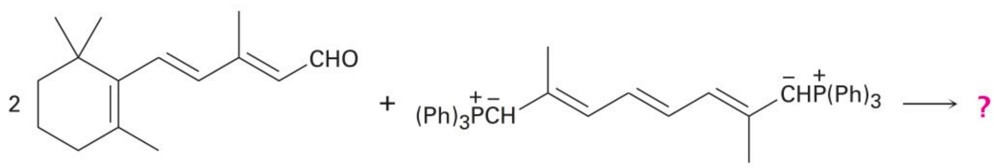
22. This hydroxyketone shows no peaks in its **IR spectrum** around 1700 cm^{-1} (for carbonyl group), but it does show broad absorption around 3400 cm^{-1} (for hydroxyl group). Justify. **(1)**



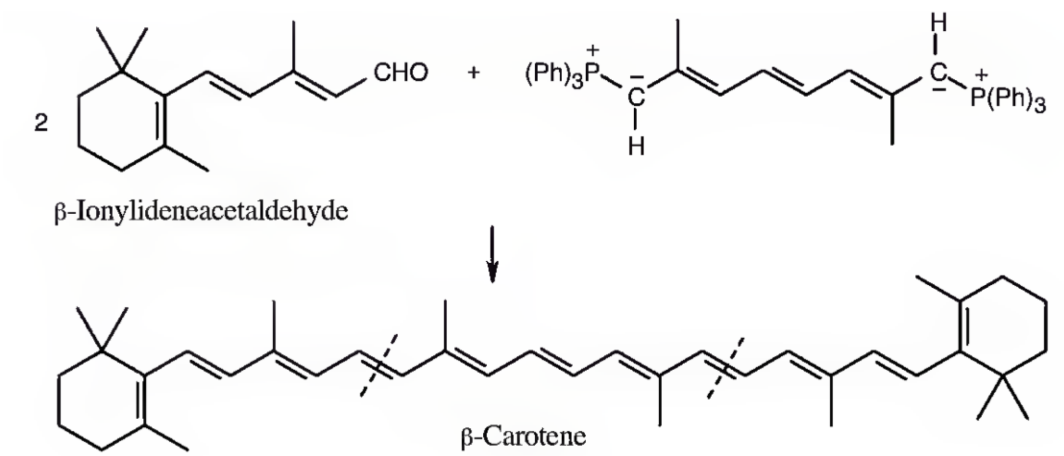
This is because of the intramolecular hemiacetal formation which will not show any IR band for the C=O group but will show the IR peak for the -OH group.



23. *β*-Carotene, a yellow food-coloring agent and dietary source of vitamin A can be prepared by a *double* **Wittig reaction** between 2 equivalents of *β*-ionylideneacetaldehyde and a *diylide.* Show the structure of the *β*-carotene product with appropriate **stereochemistry.** **(1)**



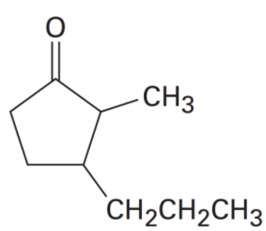
***β*-Ionylideneacetaldehyde** **A diylide**



Double Wittig Reaction-Structure is enough, mechanism is not required

24. Answer the following questions**:** **(1x2=2)**

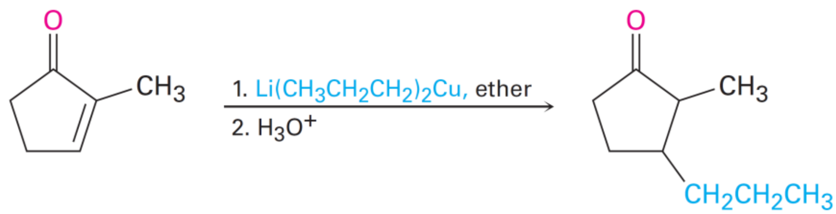
a) How might you use a **Michael addition** reaction to prepare 2-methyl-3-propylcyclopentanone?



2-Methyl-3-propylcyclopentanone

A ketone with a substituent group in its β position might be prepared by a conjugate addition of that group to an α,β-unsaturated ketone. In the present instance, the target molecule has a propyl substituent on the β carbon and might therefore be prepared from 2-methyl-2-cyclopentenone by reaction with lithium dipropylcopper.

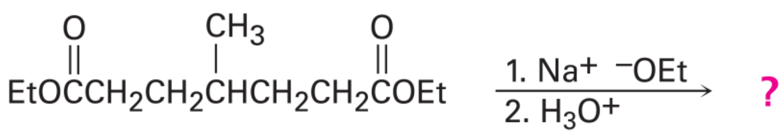
Solution

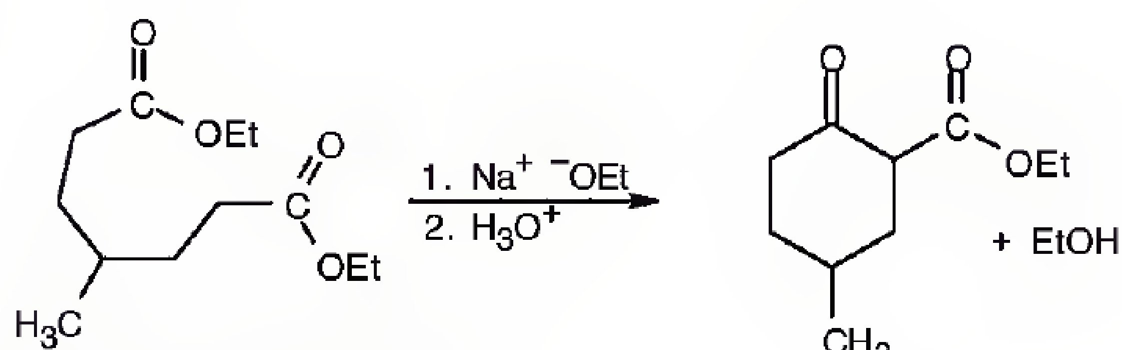


2-Methyl-2-cyclopentenone **2-Methyl-3-propylcyclopentanone**

Grignard reagent/Alkyl lithium in presence of Cu will be suitable for 1,4-addition.

b) What product would you expect from the following reaction? **Is this a named reaction**?



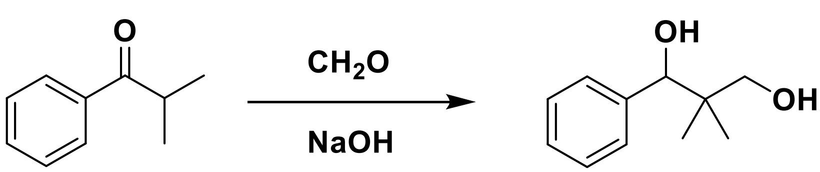


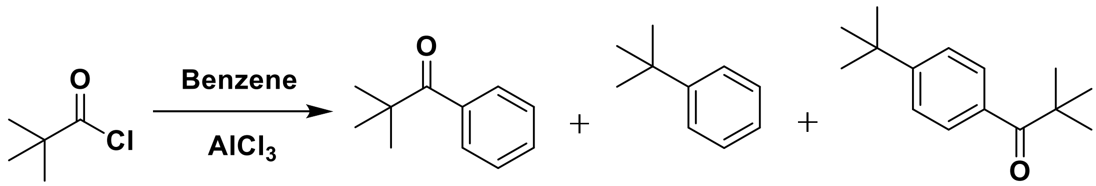
Diethyl 4-methylheptanedioate

This is Dieckmann cyclization/condensation

25. Propose suitable mechanisms for the following reactions and identify the **name reaction**. **(6)**

a)

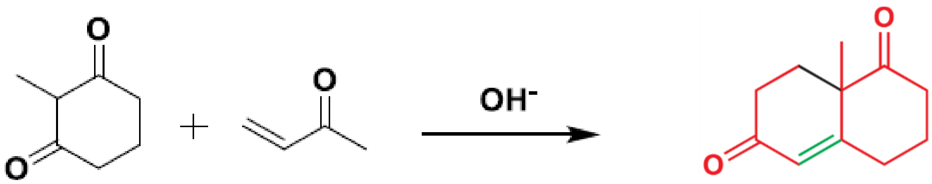




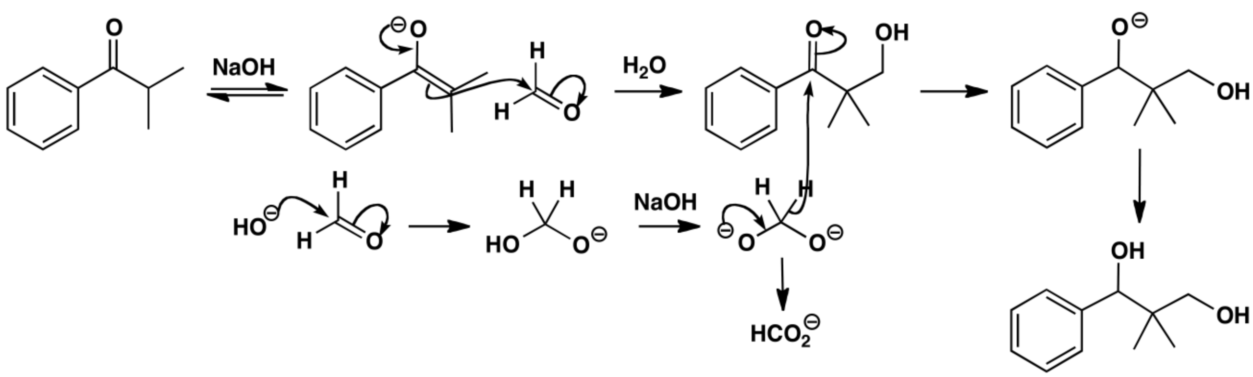
A B C

b)

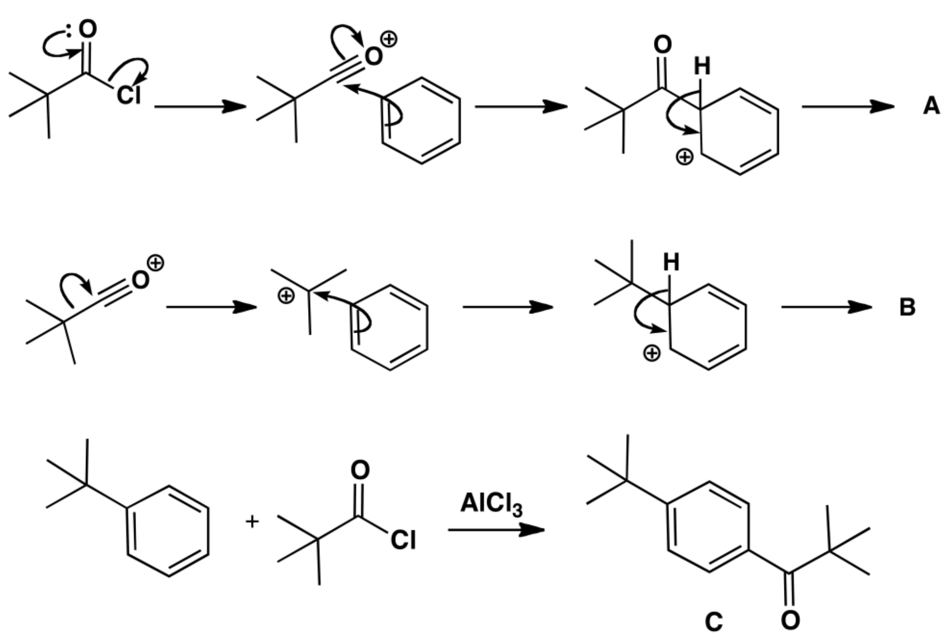
c)



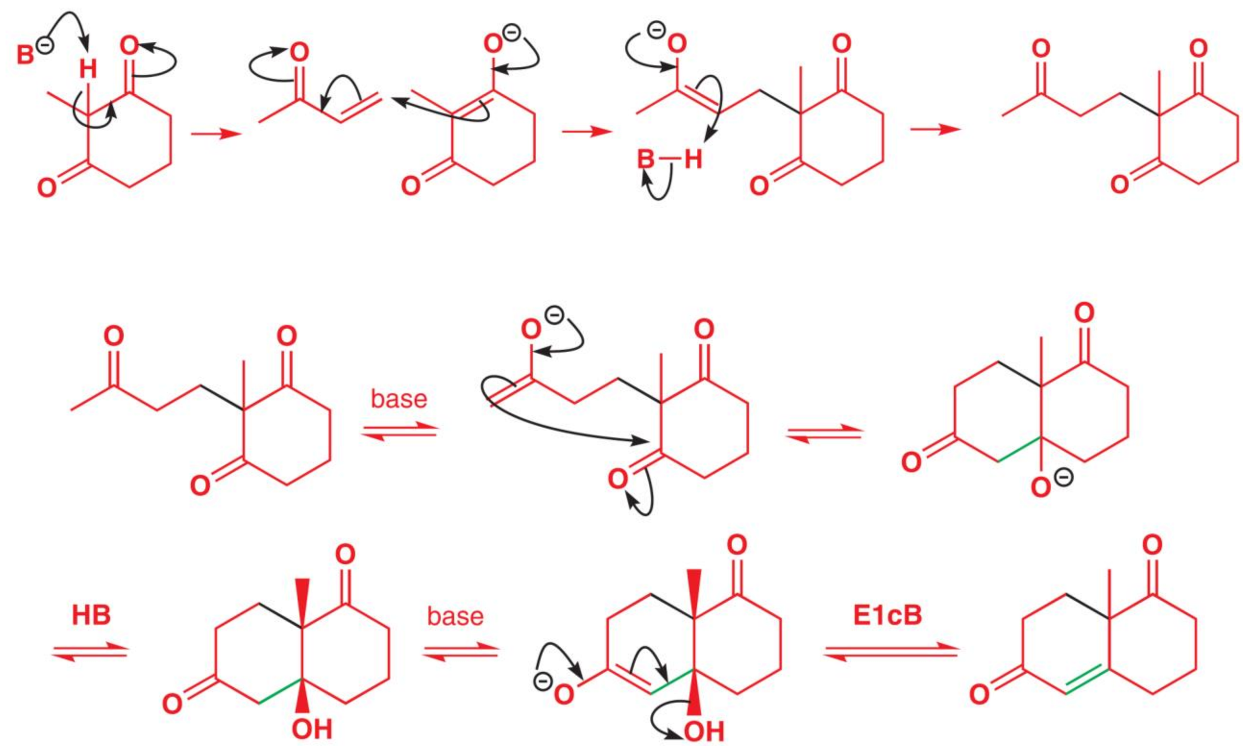
a) **Mechanisms are below, stepwise marking will be awarded for the attempt**



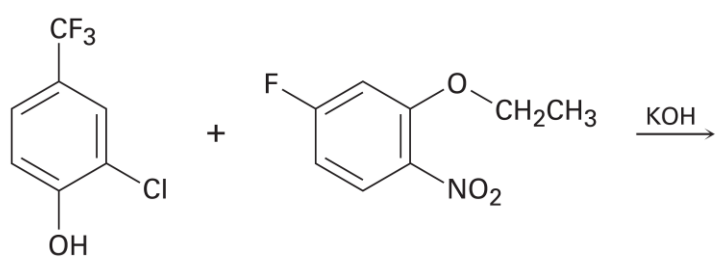
b)

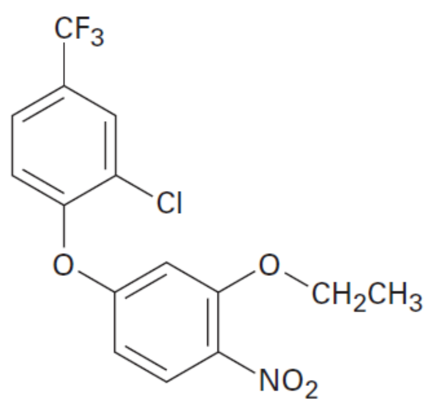


c)



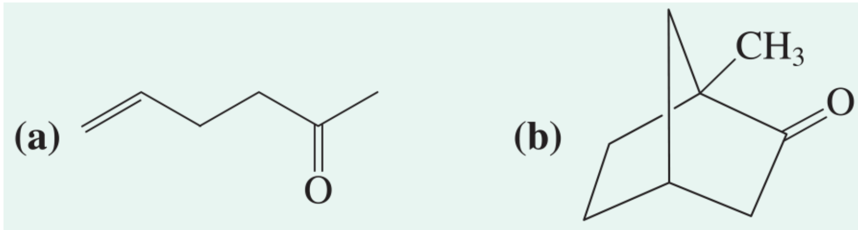
26. What is the product of the following reaction? **What kind of reaction is this?** **(1)**





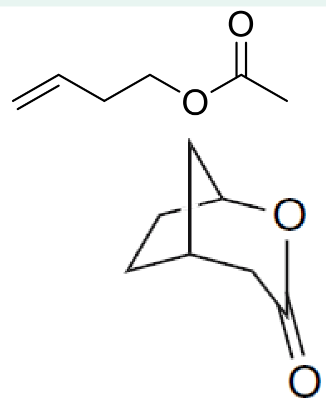
This is nucleophilic aromatic substitution (S_NAr)-Addition-elimination mechanism

27. Predict the outcome of oxidation of the following compounds with a **peroxycarboxylic acid.** **(1)**



a)

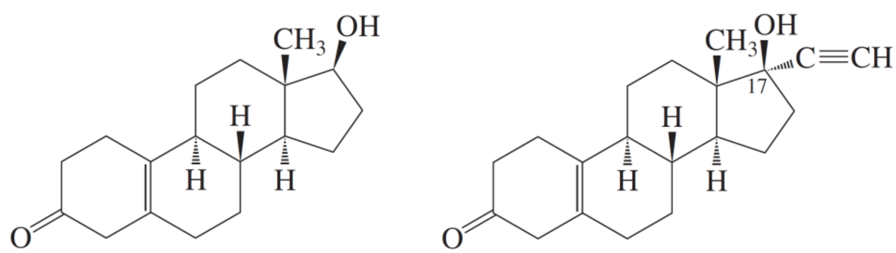
b)



Consider the methyl group on the bridgehead, product will contain the methyl group

(These are examples of Baeyer-Villiger reaction, only product structure is enough)

28. Propose a synthesis of **norethynodrel** using the starting compound (left-hand structure). [Hint: You can use acetylene, ethylene glycol, pyridinium chlorochromate as some of the reagents]. (**2**)



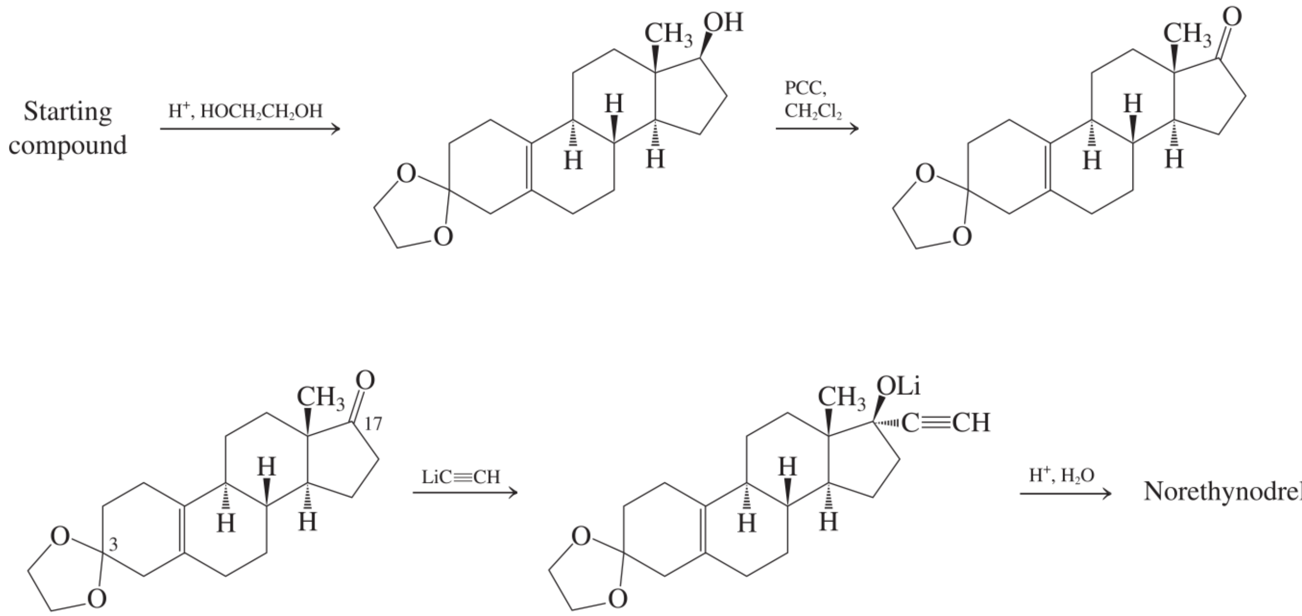
Starting compound

Norethynodrel

Step 1: Protection of the carbonyl group using ethylene glycol (to make an acetal)

Step 2: Oxidation of the 2-degree alcohol group to keto group using PCC

Step 3: Nucleophilic addition from acetylide, followed by hydrolysis of acetal using acid



Starting compound
H+, HOCH2CH2OH
CH3
OH
H
H
H
O
O
PCC,
CH2Cl2
CH3
O
17
3
LiC≡CH
OLi
C≡CH
H+, H2O
Norethynodrel

(1) Gd^{+3} with L-S coupling

$4f^7 \Rightarrow$ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ | ↑ |

$L = 0$ for half-filled 4f

$S = \frac{7}{2}$

Term symbol – $^8S_{\frac{7}{2}}$

$\mu = g\sqrt{J(J+1)}$ BM

$\mu = 2\sqrt{\frac{7}{2} \times (\frac{7}{2}+1)}$ BM since $g = 2$ for S

$\mu = \sqrt{63}$ BM $= 7.93$ BM

(2) $CoCl_4^{+2}$ have Td geometry which is spin allowed and Laporte partially allowed. Hence lower ϵ value.

$[Co(H_2O)_6]^{+2}$ – have Oh geometry is spin allowed and Laporte allowed, have higher ϵ value

(3) $[NiBr_4]^{-2}$ – Td

↑↓ ↑ ↑ d_{xz} d_{yz} d_{xy}

↑↓ ↑↓
d_{z^2} $d_{x^2-y^2}$

Paramagnetic

2 unpaired e^- makes ~~At~~ it ~~Diamagnetic~~

$[Pd\,Br_4]^{-2} \Rightarrow$ Sq. plannar

$\uparrow$

— $d_{x^2-y^2}$

$\uparrow\downarrow$ d_{xy}

$\uparrow\downarrow$ d_{z^2}

$\uparrow\downarrow$ $\uparrow\downarrow$

d_{xz} d_{yz}

No unpaired e^-
so diamagnetic

Ans-4 (a) In terms of Δ_o $[Cr(OH_2)_6]^{+3} > [Cr(OH_2)_6]^{+2}$

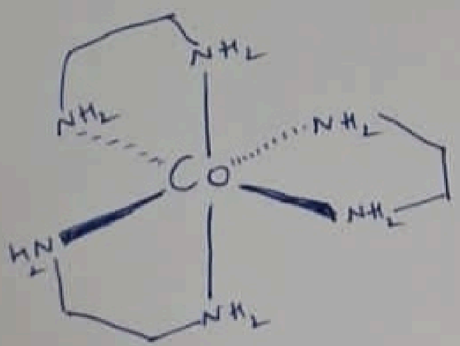
High OS favour high splitting for the same metal.

(b) $[Cr(NH_3)_6]^{+3} > [Cr(F_6)]^{-3}$

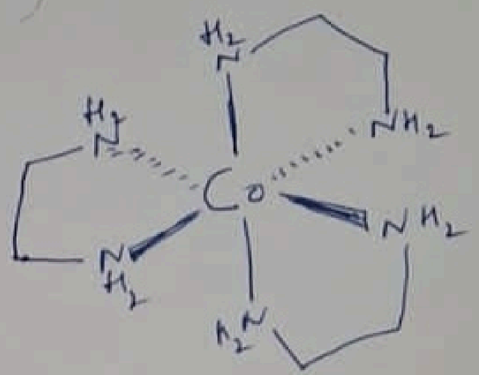
Strong field ligand favour high splitting.

4)

5) Draw the isomers of tris(ethylene diamine) Cobalt(III) Chloride assign (Δ & Λ)



Λ

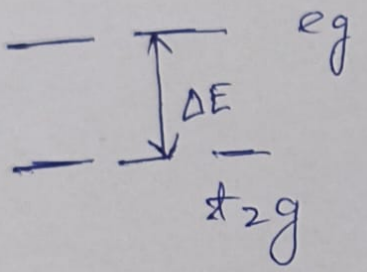


Δ

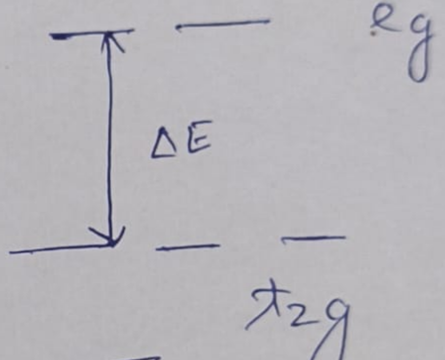
Ans-6 π donor – O^{-2}, F^-, RO^-, Cl^- etc

π acceptor – CO, CN, bpy etc

π acceptor increases the splitting energy and widens the gap b/w t_{2g} and e_g levels.



π donor



π acceptor